

MTU_Value*Service* **Technical Documentation**

**Data protocols
J1939 and CANopen**

**for GenoLine LOP 10
(Engine management systems
MDEC and ADEC/EIM)**

Application: Marine

E532303/00E



Printed in Germany

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1 Transmission values as per J1939 standard

1.1 General information

1.1.1 Message structure

A CAN message is structured as follows:

- 29 bit CAN identifier
- Statement of the length of the data field
- Up to 8 data bytes (J1939/21)

MTU also uses data packets with less than 8 data bytes, therefore an evaluation of the data field length is vital.

1.1.2 Message structure identifier

The 29 bit CAN identifier consists of the following components:

- Message priority
This is preset to 6 for all PGNs
- PGN (Parameter Group Number)
This number is structured as follows:
 - EDP – defines the structure of the CAN identifier and is always set to zero for J1939 messages
 - DP – bit for the extension of the subsequent PDU (Process Data Unit) format field
 - PF – PDU format
Field to distinguish between communication services. Together with other constituents of the PGN, it allows to describe the subsequent data field. Messages sent to all nodes (broadcast) are in PDU1 format (PDU format field > 239). Messages only sent to individual devices (peer to peer) are PDU2 format (PDU format field < 240)
 - PS – PDU Specific (DA or GE)
The definition of this data field depends on the PDU format. For PDU1 format the specific field is the target address (DA). For PDU2 format this data field contains the group extension (GE).
DA: Target address to which a message is transmitted.
GE: Forms the parameter group together with the PF.
- Source address
This is a unique device address with 8 bits (0...253).

The structure of the identifier therefore appears as follows:

28 ... 26	25	24	23 ... 16	15 ... 8	7 ... 0
Priority	PGN (Parameter Group Number)				Source address
	Res	DP	PF	DA/GE	

Abbreviations:

- DA: Destination Address
- DP: Data Page
- EDP: Extended Data Page
- GE: Group Extension
- LSB: Least Significant Bit
- PDU: Protocol Data Unit
- PF: PDU Format
- PGN: Parameter Group Number
- PS: PDU Specific
- Res: Reserved

1.1.3 PDU message structure

The maximum length of the data field is 8 bytes. However, MTU also uses shorter data fields to reduce the volume of data transmitted, or longer messages transmitted by transport protocol (e.g. DM1, DM2, TTI2, EC1, see also SAE J1939-21). Therefore the Dlc (Data length code) must be evaluated.

A request message then has, for example, the following structure (see also SAE J1939-21 Section 5.2.7.1 “Data from 0 to 8 Bytes”):

29 bit identifier						Dlc	PDU data field		
28 ... 26	25	24	23 ... 16	15 ... 8	7 ... 0		d0	d1	d2
Priority	PGN (Parameter Group Number)				Source address				
	Res	DP	PF	DA/DG			Requested PGN		

1.1.4 Nodes and baud rate

The following node and baud rate settings apply:

- SAM: Node no.: 2
- Baud rate: 250 kB
- Deterministic

Note on the process variables PV

The System Index can vary for the two applications MDEC and ADEC. The system number varies depending on the application engineering. The PV Number of the process variables comprises a System Index and a System Number. These PV Numbers vary on the basis of the information given above. Their purpose is exclusively to provide a rough orientation. The designations (e.g. engine speed, stop or start) are uniquely assigned.

1.2 SPN-PV mapping – Explanation of table

Explanation of signal ranges

The table below defines the usable signal ranges (source: SAE J1939-71, January 2008).

The value ranges summarized under "Fault indicator" have the following meanings:

- The relevant data are currently unavailable.
- Fault in sensor
- Fault in subsystem
- Fault in module

The "Unavailable" ranges are indicated when a data request is made for an inexistant signal or if the requested signal is not supported in this module.

"Not requested" summarizes ranges for which no responses are expected from the receiving device, or if receiving devices transmit a response without a request.

Range name	Numeric representation	1 byte	2 bytes	4 bytes	ASCII
Valid signal	Decimal	0 - 250	0 - 64 255	0 - 4 211 081 215	1 - 254
	Hexadecimal				
Parameter-specific information	Decimal	251	64 256 - 64 511	4 211 081 216 - 4 227 858 431	none
	Hexadecimal				
Reserved for future bits	Decimal	252 - 253	64 512 - 65 023	4 227 858 432 - 4 261 412 863	none
	Hexadecimal				
Fault indicator	Decimal	254	65 024 - 65 279	4 261 412 864 - 4 278 190 079	0
	Hexadecimal				
Unavailable or not requested	Decimal	255	65 280 - 65 535	4 278 190 080 - 4 294 967 294	255
	Hexadecimal				

Examples of table presentation

																		Operating range		
T / R	PV Index	PV Name	CAN ID MCS	Mux MCS	J1939 Name / SPN Name	SPN	PGN	S / P	K. No .	Byte Off-set	Bit Off-set	Data lengt h:	Normalization table	X-value	Y-value	Unit X	Explanation			
Example: Binary signal																				
R	PV600902	Stop Command J1939	1029	1	Auxiliary Engine Shut-down Switch	970	61441	S	10	4	5	2	Binary X-Y table	0	0		Norm range	Lower limit value		
														1	1		Norm range	Upper limit value		
														2	-2147483647		SD	Sensor Defect		
														3	2147483647		MD	Missing Data		
Example: Analog signal																				
T	PV410026	P-Lube Oil (ECU)	1001	0	Engine oil pressure	100	65263	S	2	4	0	8	Pressure 4kPa/bit	-2147483647	254	kPa	SD	Sensor Defect		
														-1	254	kPa	SD	Sensor Defect		
														0	0	kPa	Norm range	Lower limit value		
														1000000	250	kPa	Norm range	Upper limit value		
														1000001	250	kPa	Overflow			
														2147483646	250	kPa	Overflow			
														2147483647	255	kPa	MD	Missing Data		
The “SPN-PV mapping” table below only lists the values of the norm range, i.e. upper and lower limit values.																				

Explanation of table columns

• T/R	Direction (T: Transmit; R: Receive)
• PV Index	Process variable, unique number
• PV Name	Process variable, denomination
• CAN ID MCS	CAN identifier, MCS (Monitoring and Control System)
• Mux MCS	Multiplexer, MCS (Monitoring and Control System)
• J1939 Name / SPN Name	Name in J1939 protocol (may deviate from “PV Name”, as it is defined in the SAE standard)
• SPN	Suspect Parameter Number (unique identification of parameters or other elements of a control unit)
• PGN	Parameter Group Number (takes account of priority, repetition rate, data length and transmission type)
• S/P	S: PGN is standard; P: PGN is proprietary
• K. No.	Channel number
• Byte Offset	Offset (bytes)
• Bit Offset	Offset (bits)
• Data length	Number of bytes or words to be transmitted
• Normalization table	Format and scaling of expected data
• X-value	Maximum value in X-direction
• Y-value	Maximum value in Y-direction
• Unit X	Unit (X-axis)
• Explanation	(This column only applies to the sample table above.)

1.3 SPN-PV mapping – Table

T / R	PV Index	PV Name	CAN ID MCS	Mux MCS	J1939 Name / SPN Name	SPN	PGN			Byte Off-set	Bit Off-set	Data length h:	Normalization table	X-value	Y-value	Unit X
R	PV600902	Stop Command J1939	1029	1	Auxiliary Engine Shutdown Switch	970	61441	S	10	4	5	2	Binary X-Y table	0	0	
R	PV600904	Override J1939	1029	2	Engine Shutdown Override Switch	1237	65265	S	10	8	7	2	Binary X-Y table	0	0	
R	PV600901	Start Command J1939	1029	0	Engine Start	0	65325	P	10	1	1	2	Binary X-Y table	0	0	
T	PV410002	Engine Speed (ECU)	1000	1	Engine Speed	190	61444	S	2	4	0	16	Speed 0.125 rpm/bit	0	0	rpm
T	PV410026	P-Lube Oil (ECU)	1001	0	Engine Oil Pressure	100	65263	S	2	4	0	8	Pressure 4kPa/bit	0	0	kPa
T	PV410031	P-Crankcase	1001	5	Crankcase Pressure	101	65263	S	2	5	0	16	Pressure 128kPa/bit	-250000	0	kPa
T	PV410036	P-Coolant	1001	10	Coolant Pressure	109	65263	S	2	7	0	8	Pressure 2kPa/bit	0	0	kPa
T	PV410046	P-Fuel	1001	20	Engine Fuel Delivery Pressure	94	65263	S	2	1	0	8	Pressure 4kPa/bit	0	0	kPa
T	PV410049	P-Charge Air (a)	1001	23	Engine Turbocharger Boost Pressure	102	65270	S	2	2	0	8	Pressure 2kPa/bit	0	0	kPa
T	PV410056	HI Fuel Leakage	1002	5	Fuel Leakage 1	1239	65169	S	2	1	1	2	Binary X-Y table	0	0	
T	PV410057	ETC 1 Speed	1002	6	Turbocharger 1 Speed	103	65245	S	2	2	0	16	Speed 4 rpm/bit	0	0	rpm
														2570200	64255	rpm